

Financial Statement Analysis

• Basic EPS

$$\text{Basic EPS} = \frac{\text{Net Income Attributable to Common Equity Shareholders}}{\text{Weighted average number of shares outstanding}}$$

• Dilluted EPS

$$\text{Diluted EPS} = \frac{\text{Net Income Attributable to common ESH} + \text{Preferred Dividend on convertible PS} + \text{Interest (1-Tax) on convertible debt}}{\text{WANES} + \text{Equity shares on convertible securities}}$$

• Stock Options and Warrants - Treasury Method

$$\text{Number of shares to add in denominator} = \frac{(\text{Avg Market Price} - \text{Exercise Price}) \times \text{Number of shares upon conversion.}}{\text{Avg market price}}$$

• Free Cash Flow to Firm (FCFF)

$$\text{FCFF} = \text{Net Income} + \text{Interest (1-Tax)} + \text{Non-cash charges} - \text{WCInv} - \text{FCInv}$$

• Free Cash Flow to Equity (FCFE)

$$\text{FCFE} = \text{Net Income} + \text{Non cash charges} - \text{WCInv} - \text{FCInv} + \text{Net Borrowings}$$

• Relationship between FCFF and FCFE

$$\text{FCFE} = \text{FCFF} - \text{Interest paid (1-Tax)} + \text{Net Borrowings.}$$

• LIFO Reserve

$$\rightarrow \text{LIFO Reserve} = \text{FIFO inventory} - \text{LIFO inventory}$$

$$\rightarrow \Delta \text{LIFO Reserve} = \text{LIFO COGS} - \text{FIFO COGS}$$

• Average age of Asset

$$\rightarrow \text{Total Useful Life} = \frac{\text{Original Cost or Gross Block}}{\text{Annual Depreciation}}$$

$$\rightarrow \text{Average Age of Assets} = \frac{\text{Accumulated Depreciation}}{\text{Annual Depreciation}}$$

$$\rightarrow \text{Remaining Useful Life} = \frac{\text{Net PPE}}{\text{Annual Depreciation}}$$

• Defined Benefit Plan

$$\text{Fixed sum p.a} = x \% \times \text{Last Drawn Salary} \times \text{Number of years worked}$$

• Income Tax Expense

$$\text{Income Tax Expense} = \text{Tax Payable} + \Delta \text{DTL} - \Delta \text{DTA.}$$

- **Effective Tax Rate**

Effective Tax Rate = Income Tax Expense / Pre-Tax Income.

- **Cash Tax Rate**

Cash Tax Rate = Cash Taxes Paid / Pre-Tax Income.

Activity Ratios

$$\text{Receivables T/O} = \frac{\text{Revenue}}{\text{Avg Receivables}}$$

$$\text{Payables T/O} = \frac{\text{Purchases}}{\text{Avg Trade Payables}}$$

$$\text{Inventory T/O} = \frac{\text{COGS}}{\text{Avg Inventory}}$$

$$\text{Asset T/O} = \frac{\text{Revenue}}{\text{Avg Assets}}$$

$$\text{Fixed Assets T/O} = \frac{\text{Revenue}}{\text{Avg Net Fixed Assets}}$$

$$\text{Working Capital T/O} = \frac{\text{Revenue}}{\text{Avg Working Capital}}$$

→ **Days Outstanding Ratios –**

- Receivables Holding period or Day sales O/S = $(1/\text{Receivables turnover}) \times 365$
- Inventory Holding Period = $(1/\text{Inventory turnover}) \times 365$
- Number of days Payables = $(1/\text{Payables turnover}) \times 365$.

$$\text{Cash Conversion Cycle} = \text{Receivables Holding period} + \text{Inventory Holding period} - \text{Payables holding period}$$

Profitability Ratios

$$\text{Net Profit Margin} = \frac{\text{Net Profit}}{\text{Revenue}}$$

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Revenue}}$$

$$\text{Operating Profit Margin} = \frac{\text{Operating Income}}{\text{Revenue}}$$

$$\text{Pre-Tax Margin} = \frac{\text{Pre-Tax Profit}}{\text{Revenue}}$$

$$\text{Return on Assets} = \frac{\text{Net Income} + \text{Interest (1-Tax)}}{\text{Avg Total Assets}}$$

$$\text{Operating ROA} = \frac{\text{Operating Income}}{\text{Avg Total Assets}}$$

$$\text{Return on Invested Capital} = \frac{\text{After Tax Operating Profit}}{\text{Avg Long Term Capital}}$$

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Avg Total Equity}}$$

$$\text{Return on Common Equity} = \frac{\text{Net Income} - \text{Preferred Dividend}}{\text{Avg Common Equity}}$$

Liquidity Ratios

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

$$\text{Cash Ratio} = \frac{\text{Cash} + \text{Marketable Securities}}{\text{Current Liabilities}}$$

$$\text{Quick Ratio} = \frac{\text{Cash} + \text{Marketable Securities} + \text{Receivables}}{\text{Current Liabilities}}$$

$$\text{Defensive Interval Ratio} = \frac{\text{Cash} + \text{Marketable Securities} + \text{Receivables}}{\text{Average daily cash expenditures}}$$

Solvency Ratios

$$\text{Debt : Equity} = \frac{\text{Total Debt}}{\text{Shareholders' Equity}}$$

$$\text{Debt : EBITDA} = \frac{\text{Debt}}{\text{EBITDA}}$$

$$\text{Debt : Capital} = \frac{\text{Total Debt}}{\text{Total Debt} + \text{Shareholders Equity}}$$

$$\text{Financial Leverage Ratio} = \frac{\text{Average assets}}{\text{Average Equity}}$$

$$\text{Debt : Asset} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

$$\text{Interest Coverage Ratio} = \frac{\text{Earnings Before Interest \& Tax (EBIT)}}{\text{Interest Payments}}$$

$$\text{Fixed charge coverage} = \frac{\text{EBIT} + \text{lease payments}}{\text{Interest} + \text{lease payments}}$$

$$\text{Coverage ratios} = \frac{\text{Earnings available for distribution}}{\text{Relevant liability}}$$

• DuPont Analysis

→ 3 Part Dupont –

$$\text{ROE} = \frac{\text{Net Income}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Avg. Assets}} \times \frac{\text{Avg. Assets}}{\text{Avg. Equity}}$$

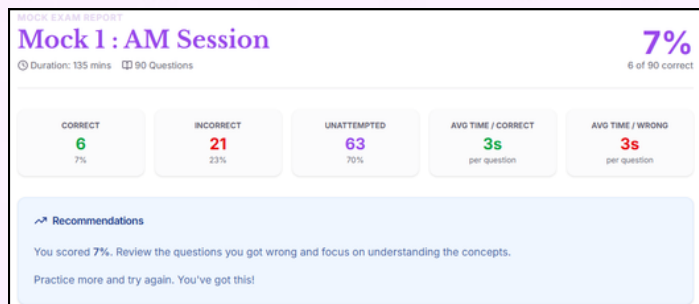
→ 5 Part Dupont –

$$\text{ROE} = \frac{\text{Net Income}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Avg. Assets}} \times \frac{\text{Avg. Assets}}{\text{Avg. Equity}}$$

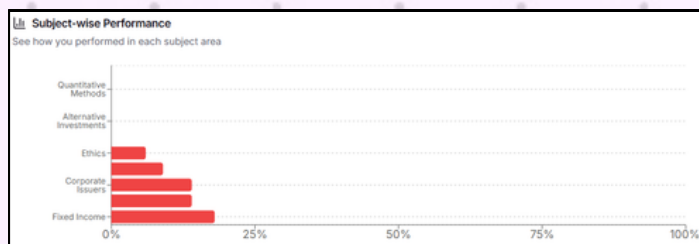
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Subject	Questions	Correct	Wrong	Skipped	Accuracy
Financial Statement Analysis	11	0	2	9	0%
Quantitative Methods	7	0	3	4	0%
Derivatives	6	0	1	5	0%
Alternative Investments	8	0	1	7	0%
Portfolio Management	6	0	0	6	0%
Ethics	16	1	6	9	6%
Equity Investments	11	1	3	7	9%
Corporate Issuers	7	1	1	5	14%
Economics	7	1	1	5	14%
Fixed Income	11	2	3	6	18%



Mock 2 : PM Session ⌚ 2:14:11

Question 4 of 90

A company accounts for Inventory using FIFO method. In Year 1, it purchases 1000 units @\$10/unit. It sells 800 units in Year 1. In year 2, it purchases 700 units @\$11/unit and sells 750 units. What is the value of closing inventory in Year 2?

☐ \$1,650

☐ \$1,617

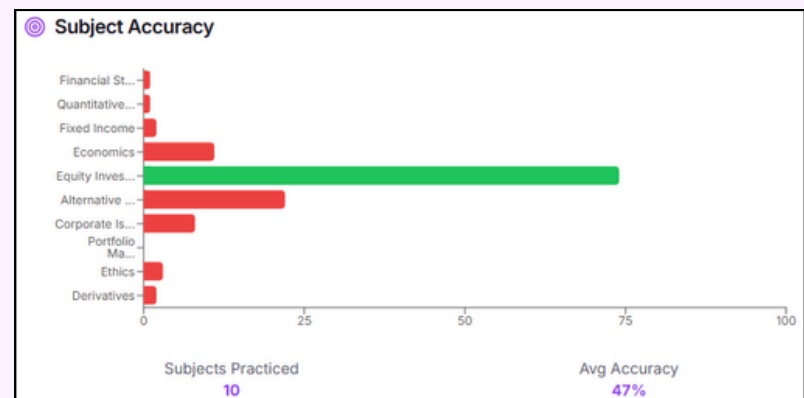
☐ \$1,500

Question Navigator

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30
31	32	33	34	35
36	37	38	39	40
41	42	43	44	45
46	47	48	49	50
51	52	53	54	55
56	57	58	59	60
61	62	63	64	65
66	67	68	69	70

< Previous ☐ Mark for Review Skip Question Next >

You can **"Skip Question"** or **"Mark for Review"**



Fixed Income

• Pricing a bond between Coupon Dates

- Full Price = Value of bond on last coupon date $\times (1 + \text{yield})^{n/N}$
- Accrued Interest =
$$\frac{\text{Coupon Amount} \times \text{number of days since the last coupon was paid}}{\text{Total number of days between two coupon dates.}}$$
- Flat price = Full price – Accrued Interest.

• Matrix Pricing – Spread based variation

Credit spread $\approx$ Yield on Corporate Bond - Yield on Benchmark Security
(where both bonds are of same maturity)

• Effective Annual Yield

$$\text{EAY} = (1 + \text{stated YTM} / n)^n$$

• Current Yield

$$\text{Current Yield} = \frac{\text{Annual cash coupon payment}}{\text{Bond Price}}$$

• Simple Yield

$$\text{Simple Yield} = \frac{\text{Annual cash coupon payment} + \text{amortized discount} - \text{amortized premium}}{\text{Bond Price}}$$

• Callable Bond

$$V_{\text{callable Bond}} = V_{\text{straight Bond}} - V_{\text{call option}}$$

• Putable Bond

$$V_{\text{putable Bond}} = V_{\text{straight Bond}} + V_{\text{put option}}$$

• Valuing Annual Coupon Paying bonds using Spot Rates

- Value of bond =
$$\frac{\text{Coupon}}{(1+S_1)} + \frac{\text{Coupon}}{(1+S_2)^2} + \frac{\text{Coupon} + \text{Face Value}}{(1+S_3)^3}$$
- Value of bond = Coupon $\times \sum$ discount factors + (Face Value $\times DF_3$)

• Spreads

- Benchmark Yield Spread = Yield on corporate bond – Yield on a benchmark security
- G-Spread = Yield on corporate bond – Yield on govt bond
- I Spread = Yield of bond – MRR

• Discount Yield

$$\text{Par Value} - \text{Discount} = \text{Purchase Price}$$

- **Bonds with Credit Risk**

- Coupon = MRR + Quoted Margin
- Discount Rate = MRR + Required Margin

- **Valuing semi-annual coupon paying bond using spot rates**

$$\text{Value of bond} = \frac{\text{Coupon}/2}{(1 + S_{0.5}/2)} + \frac{\text{Coupon}/2}{(1 + S_1/2)^2} + \frac{\text{Coupon}/2}{(1 + S_{1.5}/2)^3} + \frac{\text{Coupon}/2 + \text{Face Value}}{(1 + S_2/2)^4}$$

- **Macaulay Duration**

Macaulay Duration is weighted average time.

$$\text{Where, weight} = \frac{\text{PVCF}_i}{\sum \text{PVCF}}$$

- **Reinvestment Risk & Price Risk**

- Capital gain / (Capital loss) = Selling price of bond – Carrying Value of Bond
- Duration gap = Macaulay Duration – Investment Horizon.

- **Modified Duration**

- Modified duration = $\frac{\text{Macaulay Duration}}{(1 + \text{YTM})}$
- Modified duration (semi) = $\frac{\text{Macaulay Duration (semi)}}{(1 + \text{YTM}/2)}$
- % change in price of bond = -Mod Dur x ΔYTM

- **Approximate Modified Duration**

$$\text{Approximate Modified Duration} = \frac{V_1 - V_2}{2 \cdot V_0 \cdot \Delta \text{ytm}}$$

- **Money duration / Dollar Duration**

Money duration = Annual ModDur x Price of bond

- **Price Value Basis Point**

$$\text{PVBP} = \frac{V_1 - V_2}{2}$$

- **Factors Affecting Duration**

$$\text{Duration of a perpetuity} = \frac{(1 + \text{YTM})}{\text{YTM}}$$

- **Approximate Convexity**

- Approximate Convexity = $\frac{P_2 + P_1 - 2P_0}{(\Delta y)^2 \cdot P_0}$
- % change in price of bond = [-Mod Dur x Δytm] + [0.5 x Convexity x $(\Delta \text{ytm})^2$]

- **Money Convexity**

→ Money Convexity = Annual Convexity x Price of bond

→ \$ change in price of bond = Annual ModDur x Price of bond + Annual Convexity x Price of bond

- **Effective Duration**

$$\text{Effective duration} = \frac{V_1 - V_2}{2 \cdot V_0 \cdot \Delta \text{curve}}$$

- **Effective Convexity**

$$\rightarrow \text{Effective Convexity} = \frac{V_2 + V_1 - 2V_0}{(\Delta \text{curve})^2 \cdot V_0}$$

→ % change in price of bond = [-Effective Dur x Δcurve] + [0.5 x Effective Convexity x $(\Delta \text{curve})^2$]

- **Key Rate Duration**

→ The key rate duration of a cash flow = modified duration x weight in the portfolio.

→ % Change in value of portfolio = -Key rate duration x Δyield

- **Measuring Credit Risk**

→ Expected loss = Probability of Default x LGD

→ Loss severity = (1-Recovery Rate)

→ LGD (in monetary terms) = Expected Exposure x (1-Recovery Rate)

→ Expected Exposure = O/S amount of debt – Value of collateral.

Here, O/S amount of debt is accrued interest & principal O/S both.

→ Expected Credit Spread = P(Default) x (1-Recovery Rate)

→ Actual Credit Spread = Yield on Corporate Security – Yield on benchmark security of similar maturity

- **Market Factors**

→ Liquidity spread = yield at bid price – yield at ask price

→ Total spread = Liquidity spread + Credit Spread.

- **Sensitivity of bond price due to change in spread**

$$\text{Change in bond price due to change in spread} = -\text{Mod dur} \times (\Delta \text{spread}) + \frac{1}{2} \times \text{convexity} \times (\Delta \text{spread})^2$$

- **Debt Service Coverage Ratio**

$$\text{Debt Service Coverage Ratio} = \frac{\text{Net Operating Income}}{\text{Debt Service}}$$

Derivatives

- **Forward contract – Price & Value**

No arbitrage forward price = Spot Price + Interest + FV (Storage costs) – FV (Benefit of holding asset)

- **Long Forward (F^+)**

Payoff on Long Forward = Spot Price at maturity – Forward Price

- **Short Forward (F^-)**

Payoff on Short Forward = Forward Price – Spot Price at maturity

- **Forward Rate Agreements**

Net Settlement at Maturity =
$$\frac{(\text{Fixed Rate} - \text{MRR}) \times \text{period of loan} \times \text{Notional Principal}}{(1 + \text{MRR} \times \text{period of loan})}$$

- **Interest Rate Futures Price Quoting**

→ Interest Rate Futures Price = $100 - (100 \times \text{MRR})$

→ PVBP = Notional Principal $\times 0.01\% \times n/12$

- **Pricing a Swap**

Price of swap =
$$\frac{\$100 (1 - \text{last df})}{\sum df}$$

- **Put Call Parity**

$P + S = C + \text{PV}(X)$

- **Option Replication**

+P	$C + \text{PV}(X) - S$
-P	$-C - \text{PV}(X) + S$
+C	$P + S - \text{PV}(X)$
-C	$-P - S + \text{PV}(X)$
+S	$C + \text{PV}(X) - P$
-S	$-C - \text{PV}(X) + P$
$\text{PV}(X)$	$P + S - C$
$-\text{PV}(X)$	$-P - S + C$

- **Put-Call Forward Parity**

$$P - C = \frac{X - F}{(1+R_f)^t}$$

- **Hedge Ratio**

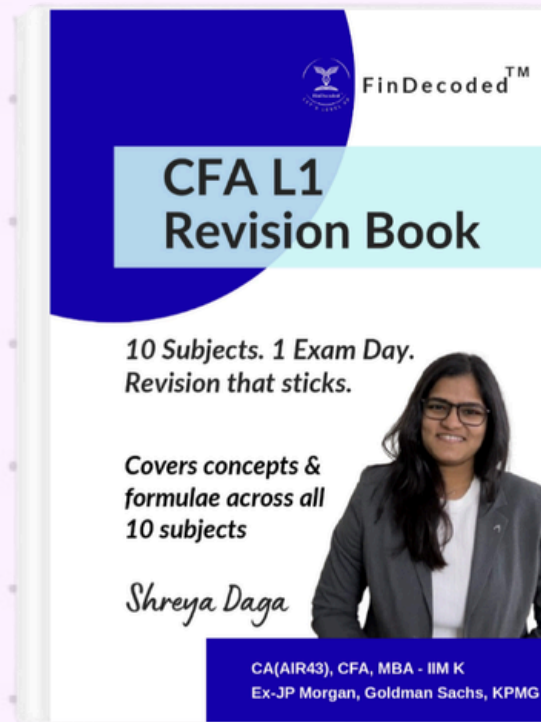
$$\text{Hedge ratio (h)} = \frac{\Delta \text{ Option}}{\Delta \text{ Stock}}$$

- **Risk Neutral Probabilities**

$$\rightarrow P(u) = \frac{1 + R_f - d}{u - d}$$

$$\rightarrow P(d) = 1 - p(u)$$





Decision Rule for Hypothesis Tests

- Be very careful when you determine whether the test is two tailed or one tailed!
- Determine which test to use (T test, F Test, Chi Square, Z test)
- Find the critical values from the relevant table by using the degree of freedom & level of significance (be careful about 1 tail / 2 tail)
- Compute the test statistic.
- Compare with the critical values; take the decision on whether or not you can reject H_0

Decision Rules

One Tailed Test

$H_0 = \text{Hypothesized Value}$

Let's say level of significance = 5%

$H_0: \mu \leq 100$

Reject H_0 if test statistic is greater than critical value (in the shaded region).

Two Tailed Test

$H_0 = \text{Hypothesized Value}$

Let's say level of significance = 5%

$H_0: \mu = 100$

Reject H_0 if test statistic is less than $-t_{\alpha/2}$ or greater than $t_{\alpha/2}$ (in the shaded regions).

Type I and Type II Errors

Decision	H_0 is True	H_0 is False
Do not reject H_0	Correct Decision	Incorrect Decision (Type II error)
Reject H_0	Incorrect Decision (Type I error)	Correct Decision

Level of significance = $P(\text{Type I error})$
 $\alpha = P(\text{Type I error})$
 $\beta = P(\text{Type II error})$

To decrease the probability of a type I error, you may think of reducing the level of significance (LOS). This will cause the below issues:-

- Increases the probability of a type II error (since by reducing the LOS, you have widened the confidence intervals, now there is a lesser chance of you rejecting the null)
- Hence, it reduces power of test.

Conversely to increase power of test, you might think of increasing level of significance. But this obviously increases the $P(\text{Type I error})$.

For a given level of significance if you want to reduce the $P(\text{Type I error})$ and the $P(\text{Type II error})$, the only way to do this is by increasing the sample size.

Capital Allocation Line

- This is a linear line, that plots expected return of portfolio against portfolio risk, by giving different weights to a risk-free asset and a risky asset.
- As per the 2-fund separation theorem, an optimal portfolio will be one which consists of some % of risky asset and some % of risk-free asset.
- While it is easy to identify what is the risk-free asset (Govt. bonds), the question is what should be the risky asset, since there are so many risky investment options.
- The risky asset is selected, based on the point of tangency (not intersection), with the efficient frontier.

It is also possible to earn more than 100% of the risky asset, by borrowing at the risk-free rate and investing it in the risky asset (we explore this further in the next chapter)

Different investors may have different CAL.

Interest Rate Arbitrage

Continuing with the above example, let's say:
 the actual 1 year forward exchange rate is INR / £ 95.
 This is lower than the theoretical forward exchange rate as implied by interest rates (103.88).
 This means that the £ is undervalued. (1 year forward £ is trading at 95 INR, versus our expectation of 103.88 INR)
 The arbitrage strategy can be constructed as below:-

Actions Today	Interest Rate	Pound	Ruppee	Relevant Ex-Rate
Borrow 1 £ for 1 year	3%	+1£		
Convert this 1£ into INR at current spot rate		-1£	+100 INR	1£ = 100 INR
Invest this 100 INR for 1 year	7%		-100 INR	
Enter into forward agreement to sell 107 INR and buy £ after 1 year				1£ = 95 INR
Total movement		0	0	

1 year later, you square off the open positions & honour your commitments

1 year later	Interest Rate	Pound	Ruppee	Relevant Ex-Rate
Investment matures and you receive 100INR + 7% interest	7%		+107 INR	
Exchange this 107 INR for £ using the forward agreement		+1.12£	-107 INR	1£ = 95 INR
Repay your borrowing of 1£ along with 4% interest.	4%	-1.04£		
Total Movement		+0.08£	0	

Thus, you make a risk free profit of 0.08£ for every pound that you had borrowed. Interest rate arbitrage states that everyone will start borrowing £, thus pushing up the demand for £, thus effectively increasing the value of £. This will keep happening till the forward rate reaches 103.88INR per £. Hence, 103.88 INR/£ can be thought of as the forward rate at which arbitrage cannot take place.

Steps for Interest rate arbitrage

- Calculate the theoretical forward rate as per interest rate differentials.
- Determine which currency is overvalued & which currency is undervalued.
- Borrow the overvalued currency.
- Invest in the undervalued currency.
- Enter into forward contract to sell (invested currency + interest)

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Doubt Solving

Stuck on a concept and don't know whom to ask?

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- Choose the **subject** you're struggling with and simply **type in your doubt**—no hesitation, no pressure.
- Our **mentor personally replies within 1-2 days** with a clear explanation.
- If the doubt is complex, we host a **live doubt-solving session** and solve it step by step with you.

The screenshot shows a web form titled "Ask a New Question". It has two main sections: "Subject" and "Your Question". The "Subject" section has a dropdown menu with "Select Subject" and a purple arrow pointing to it with the text "Choose the subject". The "Your Question" section has a text area with the placeholder "Type your doubt here clearly. You can include references to specific lecture timestamps." and a purple arrow pointing to it with the text "Enter your Doubt". At the bottom right, there are "Cancel" and "Submit Question" buttons.

The screenshot shows a question and answer interface. At the top, there's a tab "Quantitative Methods" and a green "Resolved" status. The question is: "In central limit theorem, for calculating standard error, which sample mean is being used? Since there will be sample means of at least 30 observations/samples available?". It is marked as "Asked by" a user. Below the question, there's an answer box labeled "A" that repeats the question. Below that, there's a response box labeled "A" with the text: "Standard error is the standard deviation of the distribution of sample means. We take multiple samples ($n > 30$), then calculate their means. Like this we get multiple sample means. The standard deviation of this distribution of sample means is used to compute standard error. Thus, For calculating standard error, we use distribution of multiple sample means (not individual sample means)." The response is signed "Admin Response".

Get your doubts solved directly from mentor

Corporate Issuers

Turnover Ratios

$$\text{Receivables T/O} = \frac{\text{Revenue}}{\text{Avg Receivables}}$$

$$\text{Inventory T/O} = \frac{\text{COGS}}{\text{Avg Inventory}}$$

$$\text{Payables T/O} = \frac{\text{Purchases}}{\text{Avg Trade Payables}}$$

- **Cash Conversion Cycle (CCC)**

Cash Conversion Cycle = Receivables Holding period + Inventory Holding Period
– Payables holding period

- **Working Capital**

Working Capital = Current Assets – Current Liabilities

Net Working Capital = Operating current assets – Operating current liabilities

- **Supplier Financing Rate**

Supplier Financing Rate = $[1 + a/(1+a)]^{365/(c-b)} - 1$

- **Liquidity**

Liquidity cost % = $\frac{\sum \text{Liquidity cost}}{\sum \text{FMV of assets}}$

- **Return on Invested Capital (ROIC)**

ROIC = $\frac{\text{Net Operating PAT}}{\text{Avg Invested Capital}}$

ROIC = Post Tax operating profit margin x Capital Turnover Ratio

- **Real Options**

Value of project with embedded real option = NPV + Value of Real Option – Any cost to acquire the option

- **Weighted Average Cost of Capital (WACC)**

WACC = $\{W_e \times K_e\} + \{W_d \times K_d(1-t)\}$

- **MM Proposition II : With Taxes**

$V_{\text{Levered Firm}} = V_{\text{Unlevered Firm}} + (\text{Debt} \times \text{Tax Shield})$

- **Static Trade Off Theory (MM with Taxes & Financial Distress)**

$V_{\text{Levered Firm}} = V_{\text{Unlevered Firm}} + (\text{Debt} \times \text{Tax Shield}) - \text{PV of financial distress cost}$

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Economics

- **Fiscal Multiplier**

The total increase in aggregate demand = Initial Increase in Govt Spending x Fiscal Multiplier

Where Fiscal Multiplier = $\frac{1}{1 - \text{MPC} (1 - \text{tax rate})}$

- **Balanced Budget Multiplier**

The total decrease in aggregate demand for an increase in taxes

= Initial increase in Tax (MPC) x $\frac{1}{\{1 - (\text{MPC} (1 - \text{Tax Rate}))\}}$

- **Money Neutrality**

Money Supply x Velocity = Price x Real Output

- **Real Exchange Rate**

Real Exchange Rate = Nominal Exchange Rate x $\frac{\text{CPI of base currency}}{\text{CPI of price currency}}$

Alternative Investments

- **Calculation of returns**

Multiple of invested capital: $\frac{\text{Total capital returned} + \text{Value of remaining assets}}{\text{Total capital paid during life of investment}}$

- **Use of leverage**

Leveraged portfolio return = $r \times (V_0 + V_B) - (r_B \times V_B)$

- **Return calculation for Alternative Investments**

Rate of return for an investor after fees = $\frac{V_1 - V_0 - \text{Total fees}}{V_0}$

- **Equity Interest**

Equity interest = Value of property – Value of O/S debt

- **Futures Price**

Futures price = Spot Price (1+Rf) + FV of storage costs – FV of convenience yield

Portfolio Management

• Basic Formulae

$$\text{Std Deviation} = \sqrt{\frac{(\sum x - \bar{x})^2}{n-1}}$$

$$\text{Covariance}(a,b) = \frac{\sum (R_a - \bar{R}_a)(R_b - \bar{R}_b)}{n-1}$$

$$\text{Correlation}(a,b) = \frac{\text{Covariance}(a,b)}{\sigma_a \cdot \sigma_b}$$

• Expected Return of Pf

$$ER_{pf} = (W_a)(ER_a) + W_b(ER_b)$$

• Total Risk of Pf (σ)

$$\sigma_{pf} = \sqrt{(W_a \cdot \sigma_a)^2 + (W_b \cdot \sigma_b)^2 + 2 W_a \cdot \sigma_a \cdot W_b \cdot \sigma_b \cdot r_{(a,b)}}$$

• Std Deviation pf

$$\text{std Deviation pf} = \sqrt{\frac{\text{Variance} - \text{Covariance} + \text{Covariance}}{n}}$$

• Utility function

$$U = R_{pf} - 0.5 \times \text{Var.A}$$

• Performance Evaluation

$$\rightarrow \text{Sharpe Ratio} = \frac{(ER_{pf} - R_f)}{\sigma_{pf}} \} \text{ higher the better (slope of CML / CAL)}$$

$$\rightarrow \text{Treynor Ratio} = \frac{(ER_{pf} - R_f)}{\beta_{pf}} \} \text{ higher the better (slope of SML)}$$

$$\rightarrow \text{Jensen's } \alpha = R_{pf} - \{R_f + (ER_m - R_f) \times \beta_{pf}\} \longrightarrow \text{higher the better}$$

• M-Squared Alpha

$$\rightarrow \text{M-squared Return} = \frac{R_f + (ER_{pf} - R_f) \times \sigma_m}{\sigma_{pf}}$$

$$\rightarrow \text{M-Squared Alpha} : \text{M-squared Return} - ER(\text{mkt})$$

• Return generating Models

$$\rightarrow ER_i - R_f = \text{Factor}_1(\beta_1) + \text{Factor}_2(\beta_2) + \text{Factor}_3(\beta_3) + \dots \text{Factor}_n(\beta_n)$$

$$\rightarrow \text{Market Model} : R_i = \alpha + R_m \beta + e_i$$

• Diversification Ratio

Diversification ratio = $\frac{\text{risk of an equally weighted portfolio of } n \text{ securities}}{\text{risk of a single security selected at random from the } n \text{ securities}}$

• Capital Market Line

Equation of the CML	Slope	Y intercept
$ER_{(pf)} = R_f + \frac{(ER_m - R_f) \times \sigma_{pf}}{\sigma_m}$	$\frac{(ER_m - R_f)}{\sigma_m} \rightarrow$ sharpe ratio of mkt	Risk free rate

• Securities Characterstic Line (SCL)

Slope of the SCL is beta (β)

$$\beta = \frac{\text{Covariance}_{(i,m)}}{\text{Variance}_{(m)}} = \text{Correlation}(i,m) \times \frac{\sigma_i}{\sigma_m}$$

• Securities Market Line and CAPM

Equation of the SML	Slope	Y intercept
$ER_{(pf)} = R_f + (ER_m - R_f) \times \beta$	$\frac{(ER_m - R_f)}{\beta_m} \rightarrow$ Treynor's ratio for mkt pf $(ER_m - R_f) \rightarrow$ Mkt risk premium	Risk free rate

• Return generating Models

$\rightarrow ER_i - R_f = \text{Factor}_1(\beta_1) + \text{Factor}_2(\beta_2) + \text{Factor}_3(\beta_3) + \dots \text{Factor}_n(\beta_n)$

$\rightarrow$ Market Model : $R_i = \alpha + R_m \beta + e_i$

• Real Exchange Rate

Real Exchange Rate = $\frac{\text{Nominal Exchange Rate} \times \text{CPI of base currency}}{\text{CPI of price currency}}$

• Risk of portfolio with different r ; Diversification benefit

	$r = 1$	$r = 0$	$r = -1$
Pf Risk	$w_a \cdot \sigma_a + w_b \cdot \sigma_b$	$\sqrt{(w_a \cdot \sigma_a)^2 + (w_b \cdot \sigma_b)^2}$	$w_a \cdot \sigma_a - w_b \cdot \sigma_b$
Weight for minimum variance (2 asset Pf - A & B)	There is no diversification benefit For min variance give 100% weight to asset with lower std dev.	$w_a = \frac{(\sigma_b)^2}{(\sigma_a)^2 + (\sigma_b)^2}$	$w_a = \frac{\sigma_b}{\sigma_a + \sigma_b}$

Feeling lost or unsure about how much you need to study each day?

Meet the **Momentum Meter** - In-app progress tracking

- Just enter your **target completion date**, and the app automatically calculates **days left** and **daily study hours needed**.

The screenshot shows the 'Momentum Meter' interface with the following components:

- Overall Progress:** 2%
- Current Streak:** 1 days
- Average Score:** 11%
- Latest Mock Score:** 0%
- Target Completion Date:** A section with three input fields: 'Target Date' (Aug 31, 2026), 'Days Left' (193 days), and 'Daily Target' (1h 24m/day). Below these is a progress bar showing '270h 36m remaining' and a note 'Study 1h 24m per day to meet your goal'.
- Course Progress:** A section at the bottom left.
- Access Expiry:** A section at the bottom right.

Arrows from the text below point to the 'Target Date' field, the 'Daily Target' field, and the overall 'Momentum Meter' header area.

Your personalised Momentum Meter

Enter date according your plan

get your daily study target

1:1 Onboarding Call with Mentor

When you enroll with us, you get **1:1 onboarding call with the mentor**, where we discuss your:

- **Customized study plan:** Combines your academic background, work commitments, and availability to create a realistic, balanced study schedule.
- **Structured learning & exam strategy:** Defines subject order, learning approach, revision cycles, and mock test planning for efficient preparation.
- **Progress tracking & accountability:** You receive a personalized Excel tracker to monitor topic-wise progress, revisions, and pending tasks.

Facing difficulty in only one subject?

- Don't Worry! If you are struggling with a particular subject, you can enroll in a **subject-specific course** instead of full course.
- You receive **subject-focused classes, notes, and practice questions** designed to strengthen that area effectively.

Equity Investments

• Leverage

Margin loan = 1 – Initial margin

Leverage ratio = [Assets / Equity] OR [1 / Initial margin]

Margin call price =
$$\frac{P_0 (1 - \text{Initial Margin})}{(1 - \text{Maintenance Margin})}$$

• Price Weighted Index

Formula:
$$\frac{\text{Sum of stock prices in the index}}{\text{No. of stocks in the index adjusted for splits}}$$

• Market Capitalisation Weighted Index

Formula:
$$\frac{\sum P_1 Q_1}{\sum P_0 Q_0} - 1$$

• Cost of equity, ROE, Required return concepts

→ Book value of equity = BV of assets – BV of liabilities

→ MV of equity = No of equity shares outstanding x Market price of the equity share

→ Return on Equity (ROE) =
$$\frac{\text{Net Income available to common equity shares}}{\text{Average Book Value of Equity}}$$

→ Price to book ratio =
$$\frac{\text{Mkt Value of equity}}{\text{Book Value of equity}}$$

→ Low P/B are value stocks, high P/B are growth stocks

→ Cost of equity is the minimum required rate of return for anyone to invest in your shares.

→ Capital asset pricing model: $R_f + (R_m - R_f) \times \text{Beta}$

→ Attractive investment if ROE > Cost of equity

• Costs

→ Operating Profit = [(Price – Variable costs) x Qty] – Fixed cost

→ Contribution margin = Price – Variable cost

→ Degree of Operating leverage =
$$\frac{\% \text{ change in operating profit}}{\% \text{ change in sales}} = \frac{\text{Contribution}}{\text{EBIT}}$$

→ Gross Profit = Revenue – Cost of sales

→ EBITDA = Gross profit – Operating expenses

→ EBIT = EBITDA – Depreciation & Amortization

→ Degree of Financial leverage =
$$\frac{\text{EBIT}}{\text{PBT}}$$

→ Degree of total leverage = DOL x DFL

• Return On Invested Capital (ROIC)

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Avg invested capital}} = \frac{\text{EBIT} (1 - \text{Tax})}{\text{Avg (Debt + Equity)}}$$

• Market Share

$$\text{Market share} = \frac{\text{Company's annual revenue}}{\text{Industry size}}$$

• Working Capital forecasting

$$\rightarrow \text{Accounts Receivable: } \frac{\text{DSO}}{\text{Forecasted revenues}} \times 365$$

$$\rightarrow \text{Accounts Payable: } \frac{\text{DPO}}{\text{Forecasted COGS}} \times 365$$

$$\rightarrow \text{Inventory: } \frac{\text{DOH}}{\text{Forecasted COGS}} \times 365$$

• Dividend discount models

$$\rightarrow \text{1 year holding period DDM: } V_0 = \frac{(D_1 + P_1)}{(1 + K_e)}$$

$$\rightarrow \text{Multiple-year holding period DDM: } V_0 = \frac{D_1}{(1 + K_e)} + \frac{D_2}{(1 + K_e)^2} + \dots + \frac{D_n + P_n}{(1 + K_e)^n}$$

$$\rightarrow \text{Sustainable growth rate (g) = ROE} \times \text{Retention ratio (RR)}$$

$$\rightarrow \text{Retention ratio} = 1 - \text{Dividend Payout ratio}$$

$$\rightarrow \text{Cost of equity (CAPM) = } R_f + (R_m - R_f) \times \text{Beta}$$

$$\rightarrow \text{Valuing using Gordon Growth Formula:}$$

$$\rightarrow \text{Constant growth rate: } P_0 = \frac{D_1}{(R_e - g)}$$

$$\rightarrow \text{Multi-stage growth rate: } V_0 = \frac{D_1}{(1 + K_e)} + \frac{D_2}{(1 + K_e)^2} + \dots + \frac{D_n + P_n}{(1 + K_e)^n}$$

$$\rightarrow \text{Where } P_n = \frac{D_{n+1}}{K_e - g}$$

• FCFE

$$\text{FCFE} = \text{Net income} + \text{Dep \& Amort} - \text{Increase in working capital} - \text{FC Inv} - \text{Net borrowing}$$

$$= \text{Cash Flow from Operations} - \text{FC Inv} - \text{Net borrowing}$$

• Enterprise Value

$$\text{EV / EBITDA} = \frac{\text{Enterprise Value}}{\text{EBITDA}}$$

Quantitative Methods

• Normal vs Real Rate

- $(1 + \text{Nominal } R_f) = (1 + \text{Real } R_f) \times (1 + \text{Expected Inflation})$
- $\text{Nominal } R_f \approx \text{Real } R_f + \text{Expected Inflation}$

• Holding Period Return

$$\text{HPR} = \left\{ \frac{V_1 - V_0 + \text{Intermittent CF}}{V_0} \right\} - 1$$

• Arithmetic Mean

$$\text{AM} = \frac{(R_1 + R_2 + R_3 + \dots R_n)}{N}$$

• Geometric Mean

$$\text{GM} = \sqrt[n]{(1 + R_1)(1 + R_2)(1 + R_3) \dots (1 + R_n)} - 1$$

• Harmonic Mean

$$\text{HM} = \frac{n}{\frac{1}{X_1} + \frac{1}{X_2} + \dots + \frac{1}{X_n}}$$

• Annualized return

$$\text{Annualized return} = (1 + \text{HPR})^{365/\text{days}} - 1$$

• Leveraged Return

$$\text{Leveraged Return} = \frac{r(V_0 + V_B) - r_B V_B}{V_0}$$

• TVM - Basics

- $\text{FV} = \text{PV} (1 + r)^t$
- $\text{PV} = \text{FV} (1 + r)^{-t}$
- In case of continuous compounding*
- $\text{FV} = \text{PV} \times e^{rt}$
- $\text{PV} = \text{FV} \times e^{-rt}$

• Zero Coupon Bonds

$$\text{PV} = \frac{\text{FV}}{(1 + \text{YTM})^t}$$

• Perpetual bonds

$$\text{PV of perpetual bonds} = \frac{\text{PMT}}{R}$$

• Equity Securities

$$\text{Value of preferred stock} = \frac{\text{Preferred dividend}}{K_p}$$

• Forward Interest Rate

$$(1 + S_3)^3 = (1 + S_1) (1 + 1y1y) (1 + 2y1y)$$

• Central Tendency and Dispersion

→ Sample mean = Sum of all values / n

→ Median: Mid-point of a data set when arranged in ascending / descending order

→ For odd number of observations: Median is the $\frac{(N + 1)^{\text{th}}}{2}$ observation

→ For even number of observations: It is the average of $\frac{N^{\text{th}}}{2}$ and $\frac{(N + 1)^{\text{th}}}{2}$ observation

→ Mode: Value that occurs most in the dataset

→ Trimmed Mean: Excludes a stated % of the most extreme outcomes

→ Winsorized Mean: Substituting all values after the N^{th} percentile with the value at N^{th} percentile

→ Range: Maximum value – Minimum Value

→ Mean absolute deviation: $\frac{\sum |x_i - \bar{x}|}{N}$

→ Sample variance = $\frac{\sum |x_i - \bar{x}|^2}{n - 1}$

→ Sample standard deviation = $\sqrt{\text{Sample variance}}$

→ Coefficient of variation = $\frac{\text{Std deviation of } x}{\text{Mean of } x}$

• Covariance

$$\text{Covariance: } \frac{\sum |x - \bar{x}| |y - \bar{y}|}{n - 1}$$

• Correlation

$$\text{Correlation of X and Y: } \frac{\text{Covariance of X and Y}}{(\text{Std Dev of X}) (\text{Std dev of Y})}$$

• Probability Models

→ Expected Value = Probability x Weight

$$\rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\rightarrow P(A/B) = \frac{P(A \cap B)}{P(B)}$$

• Bayes Formula

Updated probability = $\frac{\text{Probability of new info for a given event} \times \text{Prior probability}}{\text{Unconditional probability of new info}}$

OR

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

• Portfolio Mathematics

→ Variance of portfolio of 2 assets A & B = $w_a^2 \text{Var}(R_a) + w_b^2 \text{Var}(R_b) + 2 w_a w_b \text{Cov}(R_a, R_b)$

→ Correlation = $\frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}$

→ Safety first ratio = $\frac{E(R_{pf}) - R_L}{\sigma_{pf}}$

• Central Limit Theorem

Standard error of sample mean = $\frac{\sigma}{\sqrt{n}}$

• Testing if population correlation is zero

→ Pearson Correlation Coefficient: Measures the linear relationship between the variables

→ Correlation = $\frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}$

→ Spearman Rank Correlation: Non-parametric test to test whether two sets of ranks are correlated.

$$= 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

→ Test statistic (follows t-dist with $n - 2$ dof) to measure if the population correlation = 0 is:

$$\frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

• Simple linear regression

→ Linear Regression Model: $Y = b_0 + b_1 X + \text{Error term}$

→ Equation of Regression Line: $\hat{Y} = \hat{b}_0 + \hat{b}_1 X$

→ Estimated slope coefficient ($\hat{b}_1$) = $\frac{\text{Covariance}(X, Y)}{\text{Variance}(X)}$

→ Total Sum of Squares = Regression Sum of Squares + Sum of Squared Errors

→ Standard error of estimate (SEE) = $\sqrt{\text{MSE}}$

→ Coefficient of Determination (R^2) = SSR / SST

• Hypothesis test for a regression coefficient (T-test)

Test statistic =
$$\frac{\text{estimated slope coefficient} - \text{hypothesized value of slope coefficient}}{\text{Standard error of slope coefficient}}$$

• The F-Statistics

F statistic = MSR / MSE

• Predicted Values ($\hat{y}$)

$$\hat{Y} = \hat{b}_0 + \hat{b}_1 X$$

• Constructing confidence intervals for predicted values ($\hat{y}$)

Confidence interval = $\hat{y} + (\text{Critical value} \times \text{Standard error of forecast})$

• Summary of the different tests

To check	Test	Test Statistic	Degrees of freedom	Comments
Value of a population parameter (normally μ) $H_0: \mu = \mu_0$	Z test t test (for small sample sizes)	$\frac{\bar{x} - \mu_0}{\text{std error } (\bar{x})}$	N.A for z test n-1 for t test	$\bar{x}$ = sample mean μ_0 = hypothesized value of Popl ⁿ mean
Difference between means (independent Samples) $H_0: \mu_1 = \mu_2$	T test	$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_p^2}{n_1} + \frac{S_p^2}{n_2}}}$	$(n_1 + n_2 - 2)$	S_p^2 (pooled variance) = $\frac{S_1^2(n_1-1) + S_2^2(n_2-1)}{n_1 + n_2 - 2}$
Difference between means (dependent Samples) "Paired Observations Test" $H_0: \mu_1 = \mu_2$	T test	$\frac{\bar{d} - \mu_d}{\text{std error of differences}}$	n-1	We create a third distribution of d (difference between paired observations)
Value of a population variance $H_0: \sigma^2 = \sigma_0^2$	Chi Square Test	$\frac{(n-1)s^2}{\sigma_0^2}$	n-1	s^2 = Sample Variance σ_0^2 = Hypothesized Value of Popl ⁿ Variance
Comparing two population variances $H_0: \sigma_1^2 = \sigma_2^2$	F test	$\frac{S_1^2}{S_2^2}$ * keep the sample with the greater variance in the numerator	Df for numerator = $n_1 - 1$ Df for denominator = $n_2 - 1$	F test statistic > 1, Just check for the right tail.

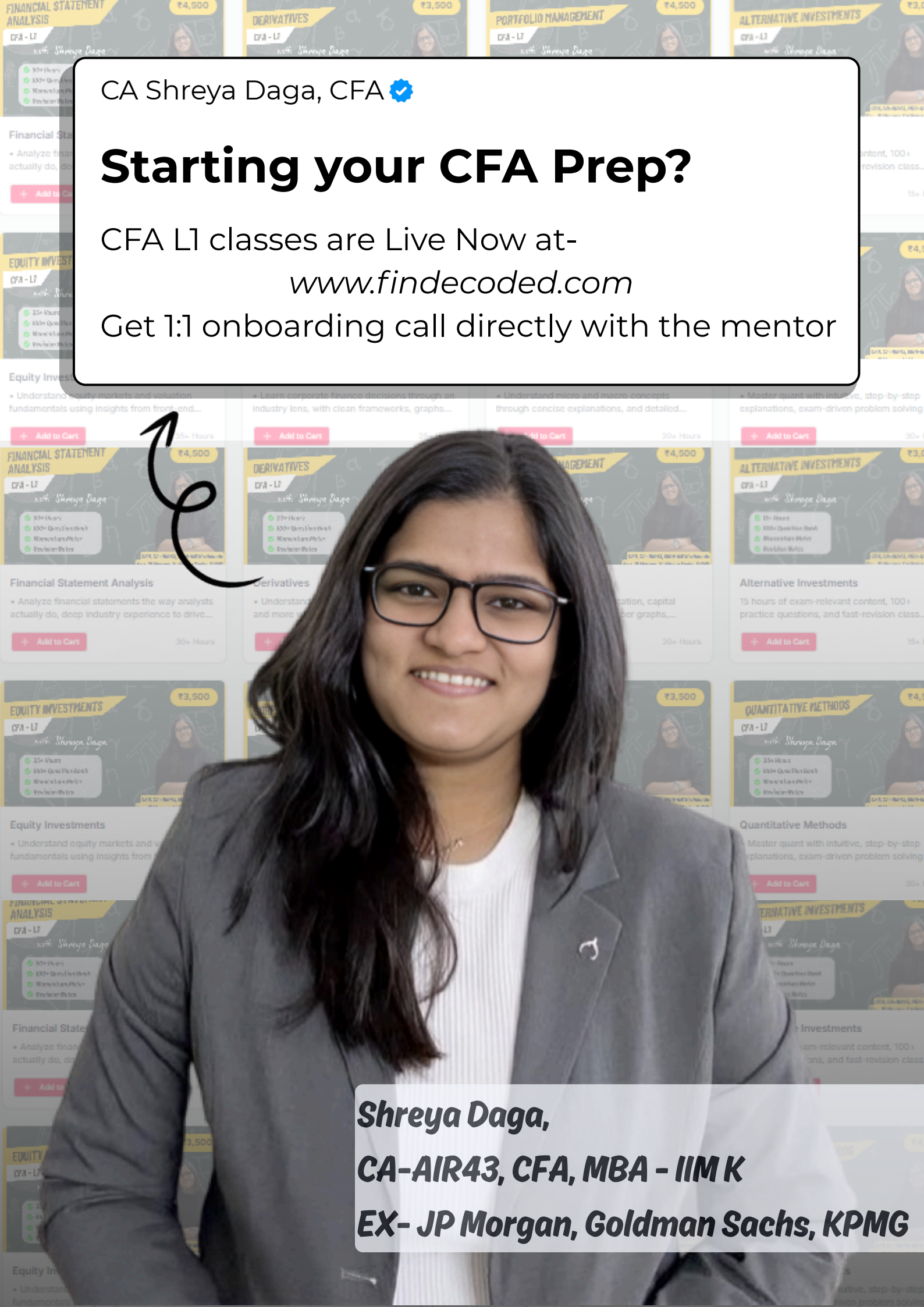
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